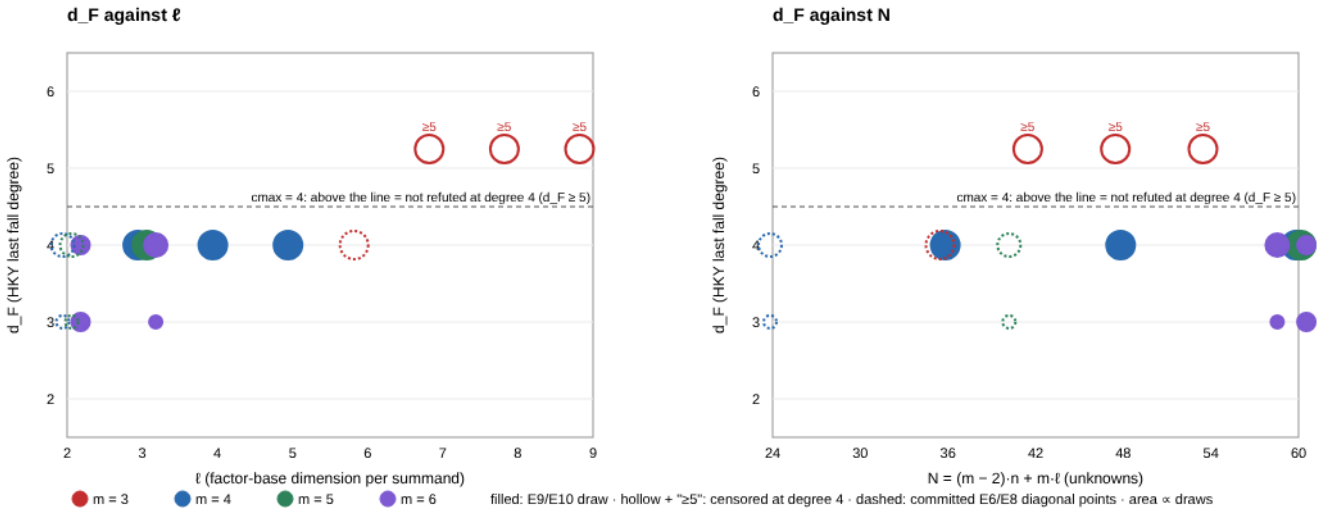
Fall-degree bounds: E9 (N = 58–60) and E10 (balanced diagonal mℓ = n). Run report

What ran. Amendment 10 of `research/fall_degree_bounds_20261008/PREREGISTRATION.md`, registered before any draw. Engine `examples/hky_par.rs` at source commit `5296ce8a7438`. Each instance built it from an S3 `git archive` (`fd-src-5296ce8.tgz`, sha256 `dade82a3…`), and the binary sha256 was `533f182b…` on all three instances. Draw rule: `ladder_sample` with `cell_seed(20261012, n, ℓ, m)`, the first 4 draws with no solutions, `cmax = 4`. A draw that is not refuted by degree 4 is recorded as `d_F ≥ 5`. Launch card `e10/launch3.card.yaml` (receipt `e10/launch_receipt3/`). Hosts: ephemeral Spot, lab account, us-east-1, AMD EPYC 9R14. Job A on m7a.8xlarge (32 vCPU); jobs B and C on m7a.4xlarge (16 vCPU). No isolation level was earned, so wall times below are indications only and are not scored. No Spot interruption.

Gates, on each instance's exact binary, all passed.

- 0: `fast_chained_count_matches_the_reference` and `chained_count` tests, 3 passed.
- 1: `e7/validate.sh`, AGREE 24 DISAGREE 0.
- 2–3: `newcells` reproduces `e6/B_5-10-2.rerun`, `e6/B_3-18-6`, `e8/E8_4-8-2` and `e8/E8_5-8-2` exactly (empty diffs).

Graph



Recorded d_F of all 36 E9/E10 draws (filled), and the committed diagonal points E6 (3, 18, 6), E6 (5, 10, 2) and E8 (4, 8, 2) (dashed), plotted against ℓ (left) and N (right). A hollow marker labelled "≥5" is a draw whose degree-4 closure stopped without 1. Marker area grows with the number of draws. Source: `plot_e9_e10.py`.

Measured values (per draw)

exp.	cell m:n:ℓ	N	on mℓ = n	draws kept	d_F per draw	degree-4 closure dim / columns	s per draw (threads)	peak RSS
E9	6:10:3	58	no	1, 2, 3, 4	4, 3, 4, 4	456,838 / 456,838 (d_F = 4 draws)	179–370 (32); 3.4 for d_F = 3	21.7 GB
E9	5:15:3	60	yes	0–3	4, 4, 4, 4	523,686 / 523,686	902–1,348 (32)	30.1 GB
E10	4:12:3	36	yes	0–3	4, 4, 4, 4	66,712 / 66,712	20–22 (16)	1.0 GB
E10	4:16:4	48	yes	0, 1, 3, 4	4, 4, 4, 4	213,053 / 213,053	477–559 (16)	5.6 GB
E10	4:20:5	60	yes	0, 1, 2, 4	4, 4, 4, 4	523,686 / 523,686	5,191–7,099 (16)	31.2 GB
E10	6:12:2	60	yes	0–3	3, 3, 4, 4	523,686 / 523,686 (d_F = 4 draws)	261–267 (16); 2.1–2.2 for d_F = 3	26.7 GB
E10	3:21:7	42	yes	0, 1, 3, 4	**≥5, ≥5, ≥5, ≥5**	117,335; 116,485; 118,185; 117,505 / 124,314	134–141 (16)	1.9 GB
E10	3:24:8	48	yes	0–3	**≥5 ×4**	186,549; 187,453; 186,323; 185,193 / 213,053	773–793 (16)	4.8 GB
E10	3:27:9	54	yes	1–4	**≥5 ×4**	160,259 (all four) / 342,541	492–533 (16)	6.7 GB

Degree-2 and degree-3 closure dimensions: `per_degree` in each record. At m = 3 they are identical across draws within a cell (62/2,338; 71/3,056; 80/3,870). Raw records: `e9/run-r3/out/E_*.jsonl`, `e10/run-r3-B/out/E_*.jsonl`, `e10/run-r3-C/out/E_*.jsonl`, with gate logs, build logs and host files beside them.

Registered readings (Amendment 10), checked by Lean `decide`

- **E9:** “bounded” (every draw ≤ 4); “grows” is false.
- **E10 m = 3 diagonal:** “grows”. Every draw of (21, 7), (24, 8) and (27, 9) is censored above 4.
- **E10 m = 4, 5, 6 diagonals:** “flat”; every cell median is ≤ 4.
- **Cross-m at fixed N:** “not rising” at N = 48 (m = 3: ≥5, m = 4: 4) and at N = 60 (4, 4, 3).

The theorem `registered_scoring` in `lean/FallDegree/Verdict.lean` is built in job E12b (log `e12/…/lean/lake.log`). Its status is recorded in RESULTS.md.

Observations (no interpretation beyond the registered readings)

- The censored draws are exactly the draws with $\ell \geq 7$. Every one of them has m = 3.
- Every draw with $\ell \leq 6$ has $d_F \leq 4$, up to N = 60.
- Where $d_F = 4$, the degree-4 closure is the whole degree-≤4 space. Where $d_F \geq 5$, it stops at 47–95% of the space without 1.
- Independent check of 3:21:7: an E11 subagent run of the same draws, on a separate host and build, also read $d_F \geq 5$ on 4/4 draws, and two solution counters agreed on 0 solutions (`e11/scaling/`). That run uses the same closure code. A separate engine, `hky_crosscheck`, was running on draw 0 when this report was written; its result is recorded in RESULTS.md when it finishes.